COMP 1405Z Fall 2026
Introduction to Computer Science I
Midterm #1 Oct. 1st, 2026

Midterm Details

You must connect to CoMaS before accessing the midterm resources and remain connected for the duration of the exam. If you suspect there are issues with your CoMaS connection, talk to a proctor or the instructor.

The midterm will be submitted through Brightspace. Put your solutions to each of the problems specified in this document in the corresponding .py file from the provided COMP1405Z-F26-Midterm1-Materials.zip file. To submit your midterm solutions, create a single .zip file with all your midterm materials and submit it to the 'COMP 1405Z Midterm #1' assignment submission on Brightspace. The official end time for the midterm is 11am (Eastern Time) but you will have until 11:10am (Eastern Time) to submit your midterm. **Do not wait until the last minute – no late submissions will be accepted**. After submitting, you should download your own zip file, extract its contents, and ensure the files have the correct content. You are not allowed to consult other students, TAs, the internet, or anybody else while completing the midterm. **You may not consult any existing code examples you have on your computer. The only materials you should access are those included in the midterm materials .zip file. You may use any IDE or text editor you generally use to write your code but must disable any intelligent auto-complete features**. You can execute your programs as many times as you want to debug and fix your code.

Problem 1 (10 marks)

Write the code for this part in the p1.py file. Write a function called sum_divisors that takes a single integer input argument N. The function must return the sum of all positive integers that evenly divide into N. That is, the sum of every number X where N divided by X produces no remainder. For example, if N is 9, the function would return 13 (1 + 3 + 9). If N was 18, the function would return 39 (1 + 2 + 3 + 6 + 9 + 18).

Problem 2 (15 marks)

Write the code for this part in the p2.py file. **You may not use additional modules, lists, dictionaries, or any other form of expandable storage other than file input/output for this problem.** Write a function called *analyze(filename)*, which accepts a single string argument representing the name of a file. You can assume the file name you are given will represent a file that contains a single positive integer value on each line. The function must return the length of the longest consecutive sequence of odd numbers inside the given file. For example, if the numbers in the file were 2 4 1 6 8 3 5 8 2 **1 3 7** 6 5, the function should return 3 because the bolded sequence is 3 odd numbers in a row and no longer sequence of odd numbers exists. A p2-tester.py file is provided to facilitate testing your function.

Problem 3 (25 marks)

Write the code for this part in the p3.py file. **You may not use additional modules, lists, dictionaries, or any other form of expandable storage other than file input/output for this problem.** In other words, you should solve this problem using only the course concepts that have been covered up to and including functions (i.e., variables, branches, loops, files, functions). **You CAN use the string.strip() method for trimming line contents**. Write a function called print_sorted_inventory_value(filename) that accepts a single string argument representing the name of a file. You can assume that the file with the given name will contain data in the following pattern representing the inventory of a business:

Item1_Name (String)
Item1_Category (String)
Item1_ItemID (integer)
Item1_Quantity (integer)
Item1_PricePerUnit (float)
Item2_Name (String)
Item2_Category (String)
Item2_ItemID (integer)
Item2_Quantity (integer)
Item2_PricePerUnit (float)
…etc…

You can assume that the Item ID values are integers and that all Item ID values within the file are unique (i.e., you can uniquely identify an item by its item ID). The total inventory value of an item can be computed by multiplying the item quantity and the price per unit. Your function must print to the console a line containing the item name, item category, and total inventory value (separated by a comma) for each item in the file

in descending order of total inventory value (i.e., highest total value to lowest total value).

Note that it may be beneficial to create additional functions and/or programmatically create files to complete this problem successfully. Remember to apply the computational thinking principles. If you cannot complete the entire problem, you can receive partial marks for successfully completing parts of the overall solution.

Example data is provided in the five inventoryX.txt files, so you can test your implementation. The problem3-expected-output.txt file contains the expected output for each of the provided inventoryX.txt files.

Submission

Create a zip containing all your midterm resources. Submit your zip to the Brightspace submission for the midterm before the cutoff time of 11:10am. No late submissions will be accepted.

Ensure you make your submission to Brightspace and check-out with a proctor or the instructor before exiting CoMaS.